

Total No. of Questions : 10]

SEAT No. :

P3256

[5461]-294

[Total No. of Pages : 3

B.E. (Information Technology)

DISTRIBUTED SYSTEM

(2012 Course) (Semester - II) (End Sem.) (414461)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) *Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8, Q9 or Q10.*
- 2) *Figures to the right indicate full marks.*
- 3) *Neat diagrams must be drawn wherever necessary.*
- 4) *Assume suitable data, if necessary.*

Q1) a) If a client and a server are placed far apart, we may see network latency dominating overall performance. How can we tackle this problem? **[5]**

b) List the various challenges during the construction of Distributed systems. Describe the challenges while designing of scalable distributed system. **[5]**

OR

Q2) a) Explain in brief, software and hardware service layers in Distributed Systems with neat diagram. **[5]**

b) Give the difference between Synchronous & Asynchronous Distributed System. Explain Event ordering with suitable example. **[5]**

Q3) a) What are Sockets? Explain the connection-oriented communication pattern using sockets with neat diagram. **[5]**

b) Explain two main characteristics of distributed event-based systems. **[5]**

OR

P.T.O.

Q4) a) Why MPI standard is needed? Explain MPI along with its core message primitives to support transient communication. [5]

b) Why Web Services are used? Explain SOAP and REST based Web Services in a nutshell. [5]

Q5) a) Explain the need to observe the global state of a distributed system with respect to: [8]

i) Distributed Garbage Collection

ii) Distributed deadlock detection

iii) Distributed termination detection

b) Explain in brief, the need of multicast communication and consensus in distributed system. Also explain the system model for Consensus process with neat diagram. [8]

OR

Q6) a) Explain the following essential requirements with respect to Mutual Exclusion: [8]

i) Safety

ii) Liveness

iii) Ordering

Also Mention the criteria to evaluate the performance of algorithms for mutual exclusion.

b) What is the significance of Replication? Mention the sequence of events when a client requests an operation to be performed in Passive replication. [8]

Q7) a) Explain the Coda Architecture along with the disconnected operation and replication strategy. [8]

b) What is meant by directory services? What are directory service Operations? [8]

OR

Q8) a) Explain how the following is achieved in Network File System: [8]

i) Access Transparency

ii) Location Transparency

iii) Mobility Transparency

b) What is *quality of service management* in multimedia applications? Describe the QoS manager's two main subtasks in brief. [8]

Q9) a) Explain the Record Protocol Layer and Handshake Layer with respect to SSL. [10]

b) Explain the design of the cryptographic algorithm using confusion and diffusion to conceal the content of a ciphertext block. [8]

OR

Q10)a) Explain the concepts of Resource Management and Stream Management with respect to Distributed Multimedia System. [10]

b) What do you mean by public-key Cryptography? Explain Digital Signatures with public keys. [8]

